

Background:

This project explores the implementation of Discrete Cosine Transform (DCT) using CUDA. DCT is a Fourier-related transform similar to the discrete Fourier transform. DCT is a widely used technique in signal processing and data compression, most notably in JPEG's image compression. Beyond static images, DCT powers feature extraction in machine learning, real-time video compression for streaming services like Netflix and YouTube, and medical imaging such as MRIs.

Challenge: DCT is computationally expensive with $O(N^2)$ complexity for 1D and $O(N^4)$ for naive 2D implementations. For real-world applications processing high-resolution images (e.g., 4K video has almost 8 million pixels split into 500,000+ 8×8 blocks), this becomes a severe bottleneck.

Motivation:

Despite its importance, GPU-based DCT algorithms remain understudied. Most research focuses on CPU implementations, leaving room for meaningful optimization work on modern parallel architectures.

Why This Problem?

Real-World Applications

- JPEG Compression: Every JPEG image uses DCT on 8×8 blocks
- All video codecs use DCT variants
- Machine Learning: Feature extraction in computer vision
- Signal Processing: Audio compression (MP3), spectral analysis
- Medical Imaging: CT scans, MRI processing

Performance Critical

Video encoding for streaming needs real-time DCT

4K video at 60fps = processing 498,960 blocks per frame $\times$ 60 = approx. 30M blocks/sec

Mobile devices need power-efficient DCT for video calls

Cloud services process millions of images, faster DCT = lower costs

Computational Bottleneck

Current state:

CPU implementations: Too slow for real-time HD/4K video

Existing GPU implementations: Often not optimized or closed-source

Gap between theoretical GPU capabilities and practical implementations

Related Work:

Papers:

<https://arxiv.org/pdf/2110.01172>

<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=1093941>

<https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=101322>
https://github.com/JeremieMelo/dct_cuda

While there are existing implementations of DCT, we plan to build an optimized version with different algorithms on a GPU. (The linked repo does a decent job of implementing the basic DCT, but doesn't use batch processing; $O(N^2)$ implementation).

Our project will be different since none of the existing works implement all the different, relatively new algorithms or benchmark them against one another.

Plan:

- Implement a basic version
- Optimize the basic version
- Implement Chen's algorithm and optimize
- Implement Lee's algorithm and optimize
- (Try to come up with our own implementation?)
- Extend the existing work to 3 other types of DCT.
- Benchmark